

YUVIKA MEHTA

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Professional Summary

Detail-oriented Computer Science undergraduate with hands-on experience in **Python programming**, **data automation**, and **machine learning**. Proven ability to design and implement **automated data pipelines**, build analytical tools, and translate business requirements into technical solutions. Strong quantitative and analytical skills with experience in **NLP**, **statistical modeling**, and **full-stack development**. Effective communicator with collaborative team experience across industry and research settings.

Education

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering — CGPA: 9.86/10

Jun 2023 – May 2027

Kattankulathur, Tamil Nadu

- Relevant Coursework: **Data Structures and Algorithms**, **Database Management Systems**, **Machine Learning**, **Probability and Queuing Theory**

Experience

Machine Learning Intern

JA Assure

Dec 2025 – Ongoing

Remote

- Designed an **automated data processing pipeline** for multilingual text and voice inputs, implementing efficient data extraction and transformation workflows using **Python**.
- Built an **embedding-based retrieval system** (Sentence-BERT + FAISS) to automate knowledge extraction from unstructured documents, improving data accessibility and operational efficiency.
- Developed **API integrations** and automated validation checks to ensure data quality and enable seamless system interoperability.

Machine Learning Intern

Indian Institute of Technology Kharagpur

Jun 2025 – Jul 2025

Kharagpur, West Bengal

- Analyzed **large-scale manufacturing datasets (30,000+ samples)** and developed **automated ML pipelines** for model training, evaluation, and comparison (Random Forest, XGBoost, ANN, LSTM), achieving $R^2 = 0.99$.
- Built an interactive **Streamlit dashboard** for **performance visualization**, experiment tracking, and automated reporting to support data-driven decision-making.

Generative AI Intern

Tata Steel Downstream Products Limited

Dec 2024 – Jan 2025

Kolkata, West Bengal

- Engineered an **LLM-powered automation framework** using **Python** and **NLP** to streamline procurement workflows, reducing manual review time by **20%** across 5 departments.
- Developed **ProcurAIytics**, an AI chatbot using **LangChain** that automated **SQL query generation** from natural language, enabling **3× faster data retrieval** with **40%** efficiency improvement.

Technical Projects

OncoVision – Breast Cancer Prediction System — Python, XGBoost, Random Forest, CNN — [GitHub](#)

Nov 2025

- Developed an **end-to-end ML pipeline** integrating ensemble models (XGBoost, Random Forest) with CNN-based image classification, achieving **97%+ accuracy** through systematic experimentation and validation.
- Implemented **statistical analysis** and **Explainable AI** techniques (SHAP, LIME) for model interpretability, and deployed a real-time **Streamlit web application** for interactive data visualization.

PerzonAI - AI Marketing Tool — Next.js, FastAPI, Google Gemini API, PostgreSQL — [GitHub](#)

Aug 2025

- Built a **full-stack application** with **FastAPI backend** and **Next.js frontend**, integrating **Google Gemini API** for AI-powered persona generation and conversation simulation.
- Designed **PostgreSQL database schema** and implemented **automated data export pipelines** for analytics, improving marketing campaign efficiency by **60%**.
- Developed **RESTful APIs** for seamless data flow between frontend and backend components.

Technical Skills

Programming: Python, SQL, C++, Java, JavaScript

Data & Analytics: Exploratory Data Analysis, statistical modeling, hypothesis testing, feature engineering, model validation, performance evaluation, data visualization

Machine Learning & NLP: Scikit-learn, TensorFlow, PyTorch, ensemble methods, Natural Language Processing, Large Language Models (LLMs)

Automation & Tools: LangChain, Hugging Face Transformers, NumPy, Pandas, SciPy

Backend & Databases: FastAPI, PostgreSQL, MongoDB, REST APIs

Frontend & Visualization: Streamlit, Next.js, Jupyter Notebook, Google Colab

Developer Tools: Git, VS Code, Microsoft Excel

Achievements and Certifications

Academic Excellence: Recognized in the **Top 2% of 2,000+ students** in the Computer Science and Technology Department (2024–25).

Hackathon Achievement: Won **1st Prize** at AgentForce AI Hackathon organized by Your Engineers for PerzonAI.

Certifications: Complete Web Development Bootcamp (Udemy), Programming in Java (NPTEL), Python Programming (Udemy)